

A cubic lower bound for strict Erdős–Straus solution counts at almost all primes

Brian Akaka

14 September 2026 — working preprint v0.7

Abstract

Let $R_{\Pi}(p)$ count distinct positive integer triples $x < y < z$ satisfying $4/p = 1/x + 1/y + 1/z$, with $p \mid y, z$ and $p \nmid x$. We establish the existence of fixed constants $c, C > 0$ and N_0 such that, for every integer $N \geq N_0$,

$$\#\{p \leq N : p \text{ prime, } R_{\Pi}(p) < c(\log p)^3\} \leq C \frac{N}{(\log N)^4}.$$

The argument counts an injectively decoded family of primitive rays, proves a cubic mean and a cubic centered variance over a complete period’s units, and transfers the second moment to actual primes with weighted Bombieri–Vinogradov estimates. Repeated moduli, ordered-pair multiplicities, diagonals and the removed small-prime prefix are included explicitly. The exact prime theorem has a Lean certificate with no remaining analytic hypothesis; its external Bombieri–Vinogradov source dependency is credited and pinned. The work is AI-assisted and has not received independent human specialist review. Originality remains subject to that review. This does not prove the Erdős–Straus conjecture.

1. Statement, conventions and the change from v0.6

All logarithms are natural unless a base is specified. For a prime p , put

$$R_{\Pi}(p) = \#\{(x, y, z) \in \mathbb{N}^3 : 1 \leq x < y < z, \quad \frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad p \mid y, p \mid z, p \nmid x\}. \quad (1)$$

Only actual triples are counted: not parameter descriptions, permutations or triples with equal coordinates. This set is finite. For example, $p/4 < x < 3p/4$ follows from the equation and strict order; after x is fixed, $0 < 4/p - 1/x < 2/y$ bounds y , and the equation determines z . The formal development proves finiteness directly.

Theorem 1 (cubic prime abundance). There exist real constants $c > 0$, $C > 0$ and an integer N_0 such that every integer $N \geq N_0$ satisfies

$$\#\{p \leq N : p \text{ prime, } R_{\Pi}(p) < c(\log p)^3\} \leq C \frac{N}{(\log N)^4}. \quad (2)$$

The constants are fixed independently of p and N . No numerical value for the onset N_0 , optimized coefficient c , or universal-prime conclusion is asserted here.

Using the classical estimate $\pi(N) \gg N/\log N$, (2) implies that the exceptional proportion among primes is $O((\log N)^{-3})$. Thus one fixed set of primes of relative natural density one satisfies the

cubic lower bound. “Relative” means that the denominator is $\pi(N)$, not N . Exceptions to this multiplicity bound need not be exceptions to the existence of any Erdős–Straus solution.

The following comparison concerns bounds and their asymptotic scales, not measured numbers of exceptional primes.

Claim	Published v0.6	This revision
Strict Type-II lower threshold	$c(\log p)^3 / \log \log p$	$c(\log p)^3$
Exceptional primes up to N	$O(N(\log \log N)^2 / (\log N)^3)$	$O(N / (\log N)^4)$
Certified analytic chain	Restricted earlier certificate	Entire specified cubic prime theorem

Up to fixed positive constants the lower threshold gains a factor $\log \log p$. The new exceptional upper scale is smaller by a factor $\log N(\log \log N)^2$. The proof architecture changes: the present unrough restricted family is analyzed by its centered variance over units, not by the older rough full-family positive-overlap argument. Published earlier versions remain available in the same Zenodo series.

1.1. Relation to earlier literature

Elsholtz and Tao [ET, Theorem 1.8] prove $f(p) \geq (\log p)^{(\log 3)/2 - o(1)}$ for their total solution count on a set of primes of relative density one; $(\log 3)/2 \approx 0.5493$. Their count permits permutations and equal coordinates. Since every triple counted in (1) is counted by their total count, (2) gives a stronger density-one prime lower bound even for a more restricted class of solutions. This comparison does not claim that every other conclusion of [ET] is strengthened.

Elsholtz and Planitzer [EP] obtain further lower bounds, including much larger values at infinitely many primes in prescribed reduced residue classes. Such extremal subsequence statements and a density-one prime statement address different questions; the present result does not supersede them. A complete originality assessment remains outstanding.

The new estimate is a multiplicity result, not an improvement of the classical exceptional-set estimate for primes with *zero* solutions. Nor is it a pointwise upper bound. In particular it does not settle the remaining exceptional primes or the universal Erdős–Straus conjecture.

2. The family and its strict decoder

Let $B = 2^n$, $L = \log B$, and

$$K = \lfloor L^3 \rfloor + 2, \quad D = \lfloor L^6 \rfloor.$$

Use the finite parameter family

$$K \leq a < b \leq B, \quad (a, b) = 1, \quad B^4 < u \leq B^6, \quad (3)$$

where all parameters are integers. For each parameter set

$$M = 3abu, \quad q = 4M - 1 = 12abu - 1, \quad s = 3a^2u. \quad (4)$$

Then $0 < s < M$, $s \mid M^2$, $(q, 4s) = 1$, and $q \leq Q_0 := 12B^8$. A parameter fires at an integer v when

$$q \mid v + 4s. \quad (5)$$

Thus its firing residue is $-4s \pmod{q}$, not $s \pmod{q}$. Because $4s \equiv a/b \pmod{q}$, this is also $-a/b$ in the unit group modulo q .

Lemma 2 (strict injection). If $p > Q_0$ is prime, every firing parameter gives a triple counted by $R_{\Pi}(p)$, and distinct parameters give distinct triples.

Proof. Define

$$x = \frac{pM + s}{q}, \quad y = pM, \quad z = \frac{pMx}{s}. \quad (6)$$

Since $4M = q + 1$, (5) gives $q \mid 4(pM + s)$; q is odd, so x is an integer. The relation $qx = pM + s$, together with $s \mid M^2$ and $(q, s) = 1$, shows that $s \mid Mx$. Therefore z is an integer divisible by p . The inequality $p > q$ gives $x > M > s$, so $y < z$; also $x < y$. All coordinates are positive. Reducing $qx = pM + s$ modulo p and using $0 < s < M < q < p$ proves $p \nmid x$. Direct substitution verifies the unit-fraction identity.

Conversely, $M = y/p$ and then $s = (4M - 1)x - pM$ are recovered from the triple. The reduced ratio $s/M = a/b$ recovers the coprime pair (a, b) , and $u = M/(3ab)$. This proves both parameter-to-packet and packet-to-solution injectivity. $\square$

Let $F(v)$ be the number of firing parameters at v , and let T be the total number of parameters. In particular

$$F(p) \leq R_{\Pi}(p) \quad (p > Q_0 \text{ prime}), \quad T \leq B^8. \quad (7)$$

3. Cubic mean and full unit-period variance

Let P be the least common multiple of the moduli q in (4), and average F over $(\mathbb{Z}/P\mathbb{Z})^\times$. Write

$$\mu = \mathbb{E}_* F = \sum_{(a,b,u)} \frac{1}{\varphi(q)}, \quad V = \mathbb{E}_* |F - \mu|^2. \quad (8)$$

This is a finite probability space. It is not an assumption of independence of congruences and not yet an average over actual primes.

Lemma 3 (unit moments). For all sufficiently large dyadic B ,

$$\frac{L^3}{96} \leq \mu \leq 44eL^3, \quad V \ll L^3. \quad (9)$$

Here is the analytic structure of the lemma, with the relevant finite identities and estimates identified for review. Lower reciprocal progression estimates and the positive harmonic mass of coprime rays give the lower mean. Deleting $a < K$ costs $O(L \log L)$ in the ray harmonic mass, compared with mass of order L^2 before summing u . A positive divisor expansion of $q/\varphi(q)$ and uniform progression summation give the upper mean. These estimates retain the floors in (3).

For a primitive Dirichlet character χ of conductor $d > 1$, define

$$A_{d,\chi} = \sum_{\substack{(a,b,u) \text{ in (3)} \\ d \mid q}} \frac{\chi(a)\overline{\chi(b)}}{\varphi(q)}.$$

Finite character orthogonality, or equivalently the unit CRT covariance identity, gives

$$V = \sum_{d>1} \sum_{\chi \bmod d \text{ primitive}} |A_{d,\chi}|^2. \quad (10)$$

The common factor $\chi(-1)$ from (5) has modulus one and disappears from the squared norm. Composite conductors and prime powers are included. There are no terms with $d > Q_0$, or with $(d, 6) > 1$.

3.1. Small conductors: $2 \leq d \leq D$

Expand the coprimality condition by Möbius inversion, the reciprocal totient by its positive divisor expansion, and sum u in its arithmetic progression. The primitive Pólya–Vinogradov bound and partial summation yield, with $\alpha_K = (1 + \log K)/K$,

$$\sum_{2 \leq d \leq D} \sum_{\chi \text{ primitive}} |A_{d,\chi}|^2 \ll L^2 \alpha_K^2 D \log^2(2D) + \frac{D^2 L^6}{B^8}. \quad (11)$$

With the choices in (3), this is $O(L^2(\log L)^4 + L^{18}/B^8) = o(L^3)$. The finite expansion includes the Möbius terms above the lower ray cutoff; they are not silently dropped. The certificate proves the required composite primitive-character bound and all these adapters.

3.2. Intermediate conductors: $D < d \leq B^2$

For two primitive rays (a, b) and (c, e) , character orthogonality reduces the energy to the positive collision condition $d \mid ae - bc$. An elementary weighted determinant estimate is

$$\sum_{\substack{1 \leq a, b, c, e \leq B \\ ae \neq bc}} \frac{\tau(|ae - bc|)}{abce} \ll (1 + \log B)^6. \quad (12)$$

To see the source of this bound, split the four variables into dyadic boxes and vary a coordinate of maximal scale X . The determinant has the form $vx - w$ and absolute value $O(X^2)$. Divisor pairing reduces to moduli $h = O(X)$. The congruence has at most $O(X \gcd(v, h)/h)$ solutions in the box, including its endpoint term. Summing $\gcd(v, h)/h$ costs $O(\tau(v) \log(2X))$; averaging $\tau(v)$ over its independent coordinate costs one further logarithm. There are $O((1 + \log B)^4)$ boxes. This proves (12); determinant zero is excluded before applying divisor pairing.

For determinant zero, primitive positive ordered rays must coincide, and $\sum_{a, b \geq K} a^{-2} b^{-2} = O(K^{-2})$. Weighted Cauchy in the reciprocal-totient expansion, followed by orthogonality, gives the intermediate bound

$$\ll \frac{L^9}{D} + \frac{L^3}{K^2} + \frac{L^6}{B^4} = O(L^3). \quad (13)$$

The first term uses the elementary uniform estimate $d/\varphi(d) \ll 1 + \log d$. Thus a sharper maximal-order totient theorem is not needed in this certificate. Restrictions are dropped only after the collision sum has become nonnegative, not from inside a signed character sum.

3.3. Large conductors: $B^2 < d \leq Q_0$

Distinct primitive rays cannot have $ae \equiv bc \pmod{d}$ in this range: $|ae - bc| < B^2 < d$. For a fixed ray put $A = 12ab$. Enlarging primitive characters to all characters and using $\sum_{d \mid g} \varphi(d) = g$ bounds the whole large-conductor range by

$$\sum_{(a,b)} \sum_{B^4 < u, v \leq B^6} \frac{\gcd(Au - 1, Av - 1)}{\varphi(Au - 1)\varphi(Av - 1)}. \quad (14)$$

The diagonal $u = v$ is averaged, not bounded by a worst-case totient factor. Define a nonnegative squarefree-supported multiplicative coefficient β by $\beta(\ell) = (\ell/(\ell-1))^2 - 1 \leq 6/\ell$ for primes ℓ . Then

$$(q/\varphi(q))^2 = \sum_{h|q} \beta(h), \quad \sum_{h \geq 1} \frac{\beta(h)}{h} < \infty, \quad \sum_{h \leq Q_0} \beta(h) \leq (1 + \log Q_0)^6.$$

Progression summation gives

$$\sum_{B^4 < u \leq B^6} \frac{Au - 1}{\varphi(Au - 1)^2} \ll \frac{L}{A} + \frac{L^6}{AB^4}. \quad (15)$$

Summing over the rays bounds the true diagonal by $O(L^3 + L^8/B^4)$.

For $u \neq v$, the gcd in (14) divides $u - v$. In dyadic blocks $u \asymp U$, $v \asymp V$, $U \leq V$, congruence counting gives

$$\sum_{v \asymp V, v \neq u} \gcd(Au - 1, Av - 1) \ll V\tau(Au - 1).$$

Divisor pairing, using $U \geq B^4$ and $A \leq 12B^2$, gives $\sum_{u \asymp U} \tau(Au - 1) \ll UL$. The elementary totient bound, reciprocal weights and dyadic summation yield an off-diagonal bound $O(L^5/K^2) = o(1)$. This includes unequal u, v scales. Equations (11), (13) and (14)–(15) prove (9).

The conclusion $V = O(\mu)$ here is a centered *unit-period variance* for (3). It is not the separate full-family positive-overlap $O(\mu)$ claim discussed in earlier research notes.

4. Paying for repeated moduli and every ordered pair

Let $r(q)$ count parameters in (3) with row modulus q , and set

$$H(m) = \sum_{d|m} r(d), \quad W(m) = H(m)^2.$$

Since $12abu = q + 1$, $r(q) \leq \tau(q + 1)^2$. Every parameter whose row modulus divides m is counted by $H(m)$. Thus both the row fiber at m and the number of ordered parameter pairs with lcm exactly m are at most $W(m)$. All repetitions and diagonal pairs remain included.

Lemma 4 (harmonic multiplicity budget). For every integer $Z \geq 1$,

$$\sum_{m \leq Z} \frac{W(m)^2}{m} \leq 2(1 + \log(2Z))^{32808}. \quad (16)$$

Proof outline. Finite Hölder and $\tau(dv) \leq \tau(d)\tau(v)$ bound the left side by

$$\sum_{d \leq Z} \frac{\tau(d)^3 \tau(d+1)^8}{d} \sum_{v \leq Z/d} \frac{\tau(v)^3}{v}.$$

For each fixed positive integer j , $\tau(n)^j \leq \tau_{2^j}(n)$ and $\sum_{n \leq Z} \tau_k(n)/n \leq (1 + \log Z)^k$. Cauchy on the d -sum uses powers 6 and 16 of τ ; shifting d to $d + 1$ costs at most two in the harmonic weight. The resulting logarithmic exponent is $8 + (64 + 65536)/2 = 32808$. This elementary budget requires no shifted-divisor correlation theorem. Its size affects the analytic onset, not the final power of $\log N$.

□

5. Actual prime prefixes and weighted transfer

For an integer N choose

$$B = 2^{\lfloor \log_2 N \rfloor / 100}, \quad Q_0 = 12B^8, \quad Z = Q_0^2 = 144B^{16}. \quad (17)$$

Eventually $B^{100} \leq N \leq B^{200}$, so $\log B$ is comparable to $\log N$, $Q_0 \leq N$, and $Z^4 \leq N$. The formal proof checks these integer-floor comparisons.

Let $\mathcal{P}_N = \{p : Q_0 < p \leq N, p \text{ prime}\}$ and $m_N = \#\mathcal{P}_N$. For every positive modulus m and reduced residue r define

$$E_N(m, r) = \#\{p \leq N : p \equiv r \pmod{m}\} - \frac{\pi(N)}{\varphi(m)}, \quad e_N(m) = \max_{(r, m)=1} |E_N(m, r)|. \quad (18)$$

The corresponding error for $\mathcal{P}_N$ is exactly $E_N(m, r) - E_{Q_0}(m, r)$. The elementary bound $|E_{Q_0}(m, r)| \leq Q_0$ holds also when $m > Q_0$. Consequently the small-prefix cost is paid for every row and every ordered pair; it is not omitted by assuming uniformity near zero.

Bombieri–Vinogradov, in the prefix form centered at $\pi(N)$, gives

$$\sum_{m \leq N^{1/4}} e_N(m) \ll_A \frac{N}{(\log N)^A} \quad (19)$$

for any fixed $A > 0$. The source-proved unconditional version used here is the pinned Formal-Pantheon theorem [FP], not a hypothesis of Theorem 1. An elementary bound is $e_N(m) \ll N(1 + \log N)/m$ for $1 \leq m \leq N$. Weighted Cauchy, (16), and (19) with $A = 32811$ give

$$\begin{aligned} \left(\sum_{m \leq Z} W(m) e_N(m) \right)^2 &\leq \left(\sum_{m \leq Z} e_N(m) \right) \left(\sum_{m \leq Z} W(m)^2 e_N(m) \right) \\ &\ll \frac{N^2}{(\log N)^2}. \end{aligned} \quad (20)$$

Write the resulting weighted error as $\mathcal{E}_N = \sum_{m \leq Z} W(m) e_N(m) \ll N/\log N$. The exact centered expansion, CRT compatibility and Lemma 4 imply

$$\sum_{p \in \mathcal{P}_N} (F(p) - \mu)^2 \leq m_N V + (1 + 2\mu) \mathcal{E}_N + (T^2 + 2\mu T) Q_0. \quad (21)$$

Here an incompatible pair contributes zero, whereas a compatible pair contributes its reduced residue at its actual lcm. Equation (21) is a transfer of a centered moment, not an independence approximation.

By (9), $m_N \ll N/\log N$, $\mu \leq T \leq B^8$, and (17),

$$\sum_{p \in \mathcal{P}_N} (F(p) - \mu)^2 \ll N(\log N)^2. \quad (22)$$

Indeed the small-prefix term is at most $36B^{24}$, which is absorbed at this scale. This step is where the finite unit-period estimate becomes a statement about actual primes.

6. Proof of Theorem 1

Equations (9) and (17) supply a fixed $d > 0$ with $\mu \geq d(\log N)^3$ eventually. Choose $c = d/2$. If $p \in \mathcal{P}_N$ and $R_{\Pi}(p) < c(\log p)^3$, then (7) and $\log p \leq \log N$ give $F(p) < \mu/2$. Each such prime therefore contributes more than $\mu^2/4$ to the sum in (22). The lower-tail second-moment inequality bounds their number by

$$\frac{4}{\mu^2} \sum_{p \in \mathcal{P}_N} (F(p) - \mu)^2 \ll \frac{N}{(\log N)^4}.$$

The at most Q_0 excluded primes also fit this bound, since $Q_0 \leq 12N^{8/100}$. Enlarging a fixed constant and onset gives (2) for every integer $N \geq N_0$. No distribution assertion about the remaining exceptional primes is required or obtained. $\square$

7. Formal statement, replay and trust boundary

The release’s entry point is `sources/CubicGoalCheck.lean`. Within the namespace `CubicCertificate`, it exports the expanded theorem `full_cubic_strict_typeII`. Its quantifiers are exactly $\exists c > 0 \exists C > 0 \exists N_0$ followed by $\forall N \geq N_0$, with the actual finite count in (1). The companion declarations `actual_count_definition` and `strict_solution_fields` expose the cardinality and strict equation fields.

The original accepted run freshly rebuilt 386 source modules and checked 378 unique axiom reports. Each report uses only a subset of `propext`, `Classical.choice` and `Quot.sound`. It includes 273 pinned external Bombieri–Vinogradov modules, one external-theorem audit wrapper, 101 accepted local-source copies and 11 new prefix-integration sources. The 66 reviewed warning lines remain visible. There are no admitted lemmas or unproved analytic inputs in this theorem’s closure.

The portable archive preserves all 386 Lean files byte for byte. Its README gives dependency installation, integrity checking and fresh-build commands. Each verification run uses a new empty local object directory, checks all expected theorem reports and compares complete compiler diagnostics with authenticated baselines after file-path normalization. Source hashes, import order and fresh result records are machine-readable. Official pinned compiler and upstream library caches remain trusted; neither Lean nor all of mathlib is rebuilt from source in this procedure.

The current replay guard authenticates the official Apple Silicon Mac compiler by its exact fingerprint before execution. It requires no paths on the author’s machine. Other operating systems require separately authenticated compiler fingerprints and reviewed replay adapters; a cross-platform build is not claimed here.

The pins are Lean 4.32.0, mathlib commit 81a5d257c8e410db227a6665ed08f64fea08e997, and FormalPantheon commit ffbb65c21afc8a36ace67720f1b0df1c63d26bd1. Third-party source attribution and Apache-2.0 notices are retained in the archive.

Mathematical step	Principal Lean modules
Actual finite strict count and injection	<code>ActualCount</code> , <code>StrictCubicBridge</code>
Unit Fourier identity and mean	<code>CanonicalCoefficient</code> , <code>RowFourier</code> , <code>MeanGrowth</code>
Complete three-range unit variance	<code>CutoffLimit</code> , <code>IntermediateGrowth</code> , <code>LargeGrowth</code> , <code>ConductorSplit</code> , <code>CubicVariance</code>
Row and ordered-lcm multiplicities	<code>ActualMultiplicity</code> , <code>HarmonicBudget</code> , <code>FamilyWeights</code>

Mathematical step	Principal Lean modules
Prime error, small prefix and scales	<code>PrefixCounts</code> , <code>PrefixEnvelope</code> , <code>PacketPrefix</code> , <code>PrefixScales</code>
Unconditional weighted BV	<code>PrefixWeightedCauchy</code> , <code>PrefixBV</code> , external <code>ProductionContract</code>
Centered prime moment and final count	<code>PrefixMoment</code> , <code>PrefixGlobalBounds</code> , <code>CubicPrimeTheorem</code> , <code>CubicGoalCheck</code>

Internal independent AI-agent audits checked the counting semantics, source closure, assumptions, multiplicities and retained diagnostics. These are not independent human specialist reviews. Formal checking supports the specified statement; it does not establish originality or replace human assessment of exposition and significance.

7.1. Claims deliberately outside this certificate

This revision does not certify an optimized numerical coefficient or numerical onset; a leading-constant normal order for the full solution count; later optimized or enlarged parameter families; all-integer extensions; the analytic proof of the Elsholtz–Tao first-moment upper bound; or the separate older full-family positive-overlap $O(\mu)$ route. None is an input to Theorem 1. The original universal-conjecture target remains unproved and is outside the import closure.

Acknowledgments and AI use

I thank Professor Christian Elsholtz for helpful correspondence, for directing me to his joint work with Planitzer, and for suggesting the study of normal orders. His comments helped shape the investigation of average versus typical solution counts. He has not reviewed the proofs, and this acknowledgment does not imply endorsement of the results or their originality. Responsibility for the claims and any errors rests with the author.

AI systems were used extensively in mathematical exploration, drafting, proof development, formalization and internal adversarial review. This work is presented transparently for independent human assessment, not on the authority of those systems. The Lean and mathlib contributors and the authors of FormalPantheon are credited for their foundational and analytic libraries. In particular the external Bombieri–Vinogradov formalization is not the author’s original contribution.

The manuscript is licensed CC BY 4.0. Third-party software and source notices retain their respective licenses.

References

- [ET] C. Elsholtz and T. Tao, *Counting the number of solutions to the Erdős–Straus equation on unit fractions*, Journal of the Australian Mathematical Society **94** (2013), 50–105. doi:10.1017/S1446788712000468. The density-one prime lower bound is Theorem 1.8. [Author manuscript](#), [arXiv v6](#).
- [EP] C. Elsholtz and S. Planitzer, *The number of solutions of the Erdős–Straus Equation and sums of k unit fractions*. [arXiv:1805.02945](#).
- [FP] FormalPantheon contributors, *FormalPantheon*, the BoundedGaps development, pinned commit `ffbb65c21afc8a36ace67720f1b0df1c63d26bd1`. [Source repository at the pinned commit](#). Used here for its source-proved unconditional Bombieri–Vinogradov theorem.
- [L] Lean and mathlib contributors, Lean 4 and mathlib. [Lean source](#), [mathlib source](#). Exact revisions and dependency pins accompany the verification sources.
- [A06] B. Akaka, *A proposed density-one lower bound for strict Erdős–Straus solution counts*, working preprint v0.6, 12 September 2026. doi:10.5281/zenodo.22706716. All versions: doi:10.5281/zenodo.22556476.